Chapter 3

QBIO 305 FA26

Summary of lecture slides

Probability

- A probability is a numerical quantity that expresses the likelihood of an event
- The probability of an event E is written Pr{E} or P(E)
- Pr{E} is always between 0 and 1 (inclusive)

Frequentist Interpretation of Probability

- Pr{E} = number of times even E occurs divided by number of times chance operation is repeated
- This interpretation only makes sense if the chance operation can be repeated many times

Probability rules

- Probability of an event is always between 0 and 1 (inclusive)
- The probability of all possible events is 1
- The probability an event does not happen is one minus the probability that it does happen
  - The complement of E (denoted E^c) is the even that E does not happen, $Pr\{E^c\} = 1 - Pr\{E\}$
- For any two events E1 and E2, Pr{E1 or E2} = Pr{E1} + Pr{E2} – Pr{E1 and E2}

Disjoint

- Two events E1 and E2 are said to be disjoint (or mutually exclusive) if Pr{E1 and E2} = 0
- Disjoint means ‘no overlap’

Conditional probability

- The conditional probability of E2 given E1 is:

$$\Pr(E_2|E_1) = \frac{\Pr(E_2 \; and \; E_1)}{\Pr(E_1)}$$

Multiplication rule

- For any two events E1 and E2,

$$\Pr\{E1 \text{ and } E2\} = \Pr\{E1\} \times \Pr\{E2|E1\}$$

To clarify…

- Pr{E1 and E2} = Pr{E2 and E1}
- Pr{E1 or E2} = Pr{E2 or E1}
- Pr{E1|E2} =/= Pr{E2|E1}

Independence

- Events E1 and E2 are independent if Pr{E1 and E2} = Pr{E1} x Pr{E2}
- This is equivalent to Pr{E2|E1} = Pr{E2}
- Also equivalent to Pr{E1|E2} = Pr{E1}
- In words: two events are independent if knowing that one event occurred does not change the probability of the other event occurring

Disjoint vs independent

- There are exercises to review these concepts in the lecture slides

Law of total probability

- For any two events E and F,

$$P(F) = P(E \text{ and } F) + P(E^c \text{ and } F)$$

- Using the multiplication rule,

$$P(F) = P(F|E)P(E) + P(F|E^c)P(E^c)$$

Bayes' Theorem

- For any two events E and F,

$$P(E|F) = P(F|E)P(E)/P(F)$$

- With the law of total probability,

$$P(E|F) = P(F|E)P(E)/(P(F|E)P(E) + P(F|E^c)P(E^c))$$

Random variable

- Variable that takes on numerical values that depend on the outcome of a chance operation
- Discrete random variable
  - The sum of all the histogram heights is 1
- Continuous random variable
  - The integral under the density curve is 1

Mean of a random variable

- Also known as the expected value or expectation,

$$\mu_Y = E(Y) = \sum_i y_i P(Y = y_i)$$

- The mean is about the average of the random variable distribution
- The mean of a random variable is different from the sample mean
  - Different samples have different sample means but the random variable mean will remain the same (e.g. rolling a die)

Variance and standard deviation of a random variable

$$\sigma_Y^2 = Var(Y) = \sum_i (y_i - \mu_Y)^2 P(Y = y_i)$$

- Standard deviation (σ_Y) is the square root of the variance
- The variance and standard deviation quantify the spread (or dispersion) of the distribution

Mean and Variance of continuous RV

$$\mu = \int x f(x) dx$$

$$\sigma^2 = \int (x - \mu)^2 f(x) dx$$

A few more things…

- Consider two random variables:
  - X has mean μ_X and variance σ_X^2
  - Y has mean μ_Y and variance σ_Y^2
- Mean of a + X: a + μ_X
- Mean of bX: $b\mu_X$
- Mean of X+Y: $\mu_X + \mu_Y$
- Mean of X-Y: $\mu_X - \mu_Y$
- Variance of a+bX: $b^2\sigma_X^2$
  - Standard deviation of a+bX: $b\sigma_X$
- Variance of X+Y (assuming independence): $\sigma_X^2 + \sigma_Y^2$
- Variance of X-Y (assuming independence): $\sigma_X^2 + \sigma_Y^2$

Sample statistics

- For constants a and b:

	Y	a+bY
Mean	$\bar{y}$	$a + b\bar{y}$
Median	m	a+bm
Standard deviation	s	bs
IQR	d	bd

Random variables X and Y

- Where
  - X has mean μ_X and variance σ_X^2
  - Y has mean μ_Y and variance σ_Y^2
  - a and b are constants:

	a+bX	X + Y	X – Y
Mean	$a + b\mu_X$	$\mu_X + \mu_Y$	$\mu_X - \mu_Y$
Variance	$b^2\sigma_X^2$	$\sigma_X^2 + \sigma_Y^2$ if X and Y indep.	$\sigma_X^2 + \sigma_Y^2$ if X and Y indep.

Size bias

- Examples of bias due to the size of groups being surveyed
- Example 1:
  - Asking each child how many siblings they have vs. asking each parent how many children they have
- Example 2:
  - Asking passengers vs. asking drivers how public transport is

Bernoulli RV (a.k.a. biased coin)

- p = probability of success (heads)
- 1-p = probability of failure (tails)
- P(Y=1) = p
- P(Y=0) = 1-p

Bernoulli RV

- $E(Y) = 1*P(Y=1)+0*P(Y=0)$

  $= 1*p$

  $= p$
- $Var(Y) = (1-p)^2P(Y=1)+(0-p)^2P(Y=0)$

  $= (1-p)^2p+p^2(1-p)$

  $=p(1-p)$
- $\sigma = \sqrt{p(1-p)}$

Binomial random variable

- If there are n independent trials (with success probability of p), count the number of successes
  - Or how many heads when flipping a coin
- This is the same as the sum of n Bernoulli random variables

Binomial random variable

$$Z = \sum_{i=1}^{n} Y_i \, , where \; each \; Y_i \; is \; an \; independent \; Bernoulli \; RV$$

$$E(Z) = E\left(\sum_{i=1}^{n} Y_i\right) = \sum_{i=1}^{n} E(Y_i) = \sum_{i=1}^{n} p = np$$

$$Var(Z) = \sum_{i=1}^{n} Var(Y_i) = \sum_{i=1}^{n} p(1-p) = np(1-p)$$

$$\sigma = \sqrt{np(1-p)}$$

Comments on Bernoulli and binomial

- Simplest random variable: either 0 or 1
- Binomial is used a lot:
  - Pass/fail for X number of items
  - Treatment for X number of patients

Albinism example

- Recessive – need 2 copies
- Suppose we have 2 people who are carriers (1 copy), chance that:
  - a child of theirs is albino is ¼
  - If they have 2 children:
    - P(both albino) = ¼ * ¼
    - P(neither albino) = ¾ * ¾
    - P(1st albino, 2nd not) = ¼ * ¾
    - P(1st not, 2nd albino) = ¾ * ¼
    - P(exactly 1 albino, 1 not in any order) = ¼ * ¾ + ¾ * ¼ = 2* ¼ * ¾
  - 3 children:
    - P(at least 1 albino) = 1 – P(all 3 not albino) = 1 – ¾ * ¾ * ¾
    - P(all 3 albino) = ¼ * ¼ * ¼
    - P(exactly 1 albino) = 3* ¼ * ¾ * ¾

	A	a
A	AA	Aa
a	Aa	aa

${}_nC_j$

- n choose j
- How many ways to pick j elements from n (order doesn't matter)
- ${}_nC_j = \binom{n}{j} = \frac{n!}{j!(n-j)!}$ $where\ n! = n(n-1)(n-2)\ldots 1$
- Note
  - 0! = 1
  - $\binom{n}{1} = \binom{n}{n-1}$

Binomial formula

- $P(exactly\ j\ successes) = \binom{n}{j} p^j (1-p)^{n-j}$
- $\binom{n}{j} = \frac{n!}{j!(n-j)!}$

- This makes calculating exact number of successes easy
- “j or more successes” and “j or less successes” is still difficult
- Approximation for this later on

Example binomial calculations

- 12 children
  - P(exactly 3 albino) = $\binom{12}{3}(¼)^3(¾)^9$
  - P(3 or fewer albino) = P(0 albino) + P(1 albino) + P(2 albino) + P(3 albino)

$$= \binom{12}{0}(¼)^0(¾)^{12} + \binom{12}{1}(¼)^1(¾)^{11} + \binom{12}{2}(¼)^2(¾)^{10} + \binom{12}{3}(¼)^3(¾)^9$$